INVENTORY MANAGEMENT SYSTEM

SQL QUERIES

Q1. Display the current stock of all products.

```
MariaDB [inventory_db]> SELECT product_id, name, stock_quantity
    -> FROM Product;
+------------+----------+----------------+
| product_id | name     | stock_quantity |
+------------+----------+----------------+
|          1 | Laptop   |             49 |
|          2 | Mouse    |             98 |
|          3 | Keyboard |             79 |
|          4 | Monitor  |             39 |
|          5 | Printer  |             29 |
+------------+----------+----------------+
5 rows in set (0.002 sec)
```

Q2. Generate a complete sales report showing customer, product, quantity, and total amount.

```
MariaDB [inventory_db]> SELECT
    ->     s.sales_id,
    ->     s.sales_date,
    ->     c.name AS customer_name,
    ->     p.name AS product_name,
    ->     sd.quantity,
    ->     sd.selling_price,
    ->     (sd.quantity * sd.selling_price) AS total_amount
    -> FROM Sales s
    -> JOIN Customer c ON s.customer_id = c.customer_id
    -> JOIN Sales_Details sd ON s.sales_id = sd.sales_id
    -> JOIN Product p ON sd.product_id = p.product_id;
+----------+------------+---------------+--------------+----------+---------------+--------------+
| sales_id | sales_date | customer_name | product_name | quantity | selling_price | total_amount |
+----------+------------+---------------+--------------+----------+---------------+--------------+
|        1 | 2026-02-20 | Rahul Sharma  | Laptop       |        1 |      52000.00 |     52000.00 |
|        2 | 2026-02-21 | Amit Verma    | Mouse        |        2 |        600.00 |      1200.00 |
|        3 | 2026-02-22 | Priya Singh   | Keyboard     |        1 |       1000.00 |      1000.00 |
|        4 | 2026-02-23 | Neha Gupta    | Monitor      |        1 |      13000.00 |     13000.00 |
|        5 | 2026-02-24 | Rohit Kumar   | Printer      |        1 |      17000.00 |     17000.00 |
+----------+------------+---------------+--------------+----------+---------------+--------------+
5 rows in set (0.002 sec)
```

Q3. Calculate the total revenue generated from sales.

```
MariaDB [inventory_db]> SELECT SUM(quantity * selling_price) AS Total_Revenue
    -> FROM Sales_Details;
+---------------+
| Total_Revenue |
+---------------+
|      84200.00 |
+---------------+
1 row in set (0.002 sec)
```

Q4. Calculate the total purchase cost of products.

```
MariaDB [inventory_db]> SELECT SUM(quantity * cost_price) AS Total_Purchase_Cost
    -> FROM Purchase_Detail;
+---------------------+
| Total_Purchase_Cost |
+---------------------+
|           354000.00 |
+---------------------+
1 row in set (0.002 sec)
```

Q5. Calculate the total profit of the business.

```
MariaDB [inventory_db]> SELECT
    ->     (SELECT SUM(quantity * selling_price) FROM Sales_Details) -
    ->     (SELECT SUM(quantity * cost_price) FROM Purchase_Detail)
    -> AS Total_Profit;
+--------------+
| Total_Profit |
+--------------+
|   -269800.00 |
+--------------+
1 row in set (0.001 sec)
```

Q6. Display products with low stock (less than 20 units).

```
MariaDB [inventory_db]> SELECT name, stock_quantity
    -> FROM Product
    -> WHERE stock_quantity < 20;
Empty set (0.000 sec)
```

Q7. Find the top-selling product.

```
MariaDB [inventory_db]> SELECT
    ->     p.name,
    ->     SUM(sd.quantity) AS total_sold
    -> FROM Sales_Details sd
    -> JOIN Product p ON sd.product_id = p.product_id
    -> GROUP BY p.name
    -> ORDER BY total_sold DESC
    -> LIMIT 1;
+-------+------------+
| name  | total_sold |
+-------+------------+
| Mouse |          2 |
+-------+------------+
1 row in set (0.003 sec)
```

Q8. Generate a monthly sales report.

```
MariaDB [inventory_db]> SELECT
    ->     MONTH(s.sales_date) AS month,
    ->     SUM(sd.quantity * sd.selling_price) AS monthly_revenue
    -> FROM Sales s
    -> JOIN Sales_Details sd ON s.sales_id = sd.sales_id
    -> GROUP BY MONTH(s.sales_date);
+-------+-----------------+
| month | monthly_revenue |
+-------+-----------------+
|     2 |        84200.00 |
+-------+-----------------+
1 row in set (0.004 sec)
```

Q9. Display all supplier details.

```
MariaDB [inventory_db]> SELECT *
    -> FROM Supplier;
+-------------+------------------+------------+------------------+-----------+
| supplier_id | name             | phone      | email            | address   |
+-------------+------------------+------------+------------------+-----------+
|           1 | ABC Traders      | 9876543210 | abc@gmail.com    | Delhi     |
|           2 | XYZ Distributors | 9123456780 | xyz@gmail.com    | Mumbai    |
|           3 | Global Supplies  | 9012345678 | global@gmail.com | Chennai   |
|           4 | Tech Wholesale   | 9988776655 | tech@gmail.com   | Bangalore |
|           5 | Prime Traders    | 9090909090 | prime@gmail.com  | Pune      |
+-------------+------------------+------------+------------------+-----------+
5 rows in set (0.002 sec)
```

Q10. Display all customer details.

```
MariaDB [inventory_db]> SELECT *
    -> FROM Customer;
+-------------+--------------+------------+-----------------+
| customer_id | name         | phone      | email           |
+-------------+--------------+------------+-----------------+
|           1 | Rahul Sharma | 9000000001 | rahul@gmail.com |
|           2 | Amit Verma   | 9000000002 | amit@gmail.com  |
|           3 | Priya Singh  | 9000000003 | priya@gmail.com |
|           4 | Neha Gupta   | 9000000004 | neha@gmail.com  |
|           5 | Rohit Kumar  | 9000000005 | rohit@gmail.com |
+-------------+--------------+------------+-----------------+
5 rows in set (0.001 sec)
```

Q11. Display products belonging to the Electronics category.

```
MariaDB [inventory_db]> SELECT name, category, price
    -> FROM Product
    -> WHERE category = 'Electronics';
+----------+-------------+----------+
| name     | category    | price    |
+----------+-------------+----------+
| Laptop   | Electronics | 50000.00 |
| Mouse    | Electronics |   500.00 |
| Keyboard | Electronics |   800.00 |
| Monitor  | Electronics | 12000.00 |
| Printer  | Electronics | 15000.00 |
+----------+-------------+----------+
5 rows in set (0.002 sec)
```

Q12. Display products with price greater than 10,000.

```
MariaDB [inventory_db]> SELECT name, price
    -> FROM Product
    -> WHERE price > 10000;
+---------+----------+
| name    | price    |
+---------+----------+
| Laptop  | 50000.00 |
| Monitor | 12000.00 |
| Printer | 15000.00 |
+---------+----------+
3 rows in set (0.002 sec)
```

Q13. Count the total number of products.

```
MariaDB [inventory_db]> SELECT COUNT(*) AS Total_Products
[    -> FROM Product;
+----------------+
| Total_Products |
+----------------+
|              5 |
+----------------+
1 row in set (0.002 sec)
```

Q14. Display the average price of products.

```
MariaDB [inventory_db]> SELECT AVG(price) AS Average_Price
[    -> FROM Product;
+---------------+
| Average_Price |
+---------------+
|  15660.000000 |
+---------------+
1 row in set (0.002 sec)
```

Q15. Display the maximum product price.

```
MariaDB [inventory_db]> SELECT MAX(price) AS Highest_Price
[    -> FROM Product;
+---------------+
| Highest_Price |
+---------------+
|      50000.00 |
+---------------+
1 row in set (0.001 sec)
```

Q16. Display the minimum product price.

```
MariaDB [inventory_db]> SELECT MIN(price) AS Lowest_Price
    -> FROM Product;
+---------------+
| Lowest_Price |
+---------------+
|       500.00 |
+---------------+
1 row in set (0.001 sec)
```

Q17. Display total number of customers.

```
MariaDB [inventory_db]> SELECT COUNT(*) AS Total_Customers
    -> FROM Customer;
+-----------------+
| Total_Customers |
+-----------------+
|               5 |
+-----------------+
1 row in set (0.001 sec)
```

Q18. Display sales made on a specific date.

```
MariaDB [inventory_db]> SELECT *
    -> FROM Sales
    -> WHERE sales_date = '2026-02-20';
+----------+------------+-------------+
| sales_id | sales_date | customer_id |
+----------+------------+-------------+
|        1 | 2026-02-20 |           1 |
+----------+------------+-------------+
1 row in set (0.002 sec)
```

Q19. Display products supplied by Supplier with ID = 1.

```
MariaDB [inventory_db]> SELECT name
    -> FROM Product
    -> WHERE supplier_id = 1;
+---------+
| name    |
+---------+
| Laptop |
+---------+
1 row in set (0.004 sec)
```

Q20. Display total quantity sold for each product.

```
MariaDB [inventory_db]> SELECT
    ->     product_id,
    ->     SUM(quantity) AS Total_Quantity_Sold
    -> FROM Sales_Details
    -> GROUP BY product_id;
+------------+---------------------+
| product_id | Total_Quantity_Sold |
+------------+---------------------+
|          1 |                   1 |
|          2 |                   2 |
|          3 |                   1 |
|          4 |                   1 |
|          5 |                   1 |
+------------+---------------------+
5 rows in set (0.002 sec)
```

Q21. Display all products sorted by price in descending order.

```
MariaDB [inventory_db]> SELECT name, price
    -> FROM Product
    -> ORDER BY price DESC;
+----------+----------+
| name     | price    |
+----------+----------+
| Laptop   | 50000.00 |
| Printer  | 15000.00 |
| Monitor  | 12000.00 |
| Keyboard |   800.00 |
| Mouse    |   500.00 |
+----------+----------+
5 rows in set (0.001 sec)
```

Q22. Display the names of customers in alphabetical order.

```
MariaDB [inventory_db]> SELECT name
    -> FROM Customer
    -> ORDER BY name ASC;
+--------------+
| name         |
+--------------+
| Amit Verma   |
| Neha Gupta   |
| Priya Singh  |
| Rahul Sharma |
| Rohit Kumar  |
+--------------+
5 rows in set (0.001 sec)
```

Q23. Display the total number of suppliers.

```
MariaDB [inventory_db]> SELECT COUNT(*) AS Total_Suppliers
    -> FROM Supplier;
+-----------------+
| Total_Suppliers |
+-----------------+
|               5 |
+-----------------+
1 row in set (0.001 sec)
```

Q24. Display products whose stock is between 20 and 100.

```
MariaDB [inventory_db]> SELECT name, stock_quantity
    -> FROM Product
    -> WHERE stock_quantity BETWEEN 20 AND 100;
+----------+----------------+
| name     | stock_quantity |
+----------+----------------+
| Laptop   |             49 |
| Mouse    |             98 |
| Keyboard |             79 |
| Monitor  |             39 |
| Printer  |             29 |
+----------+----------------+
5 rows in set (0.001 sec)
```

Q25. Display customers whose name starts with letter ‘R’.

```
MariaDB [inventory_db]> SELECT name
    -> FROM Customer
    -> WHERE name LIKE 'R%';
+---------------+
| name          |
+---------------+
| Rahul Sharma |
| Rohit Kumar  |
+---------------+
2 rows in set (0.001 sec)
```

Q26. Display all sales along with customer names.

```
MariaDB [inventory_db]> SELECT
    ->     s.sales_id,
    ->     s.sales_date,
    ->     c.name AS customer_name
    -> FROM Sales s
    -> JOIN Customer c
    -> ON s.customer_id = c.customer_id;
+----------+------------+---------------+
| sales_id | sales_date | customer_name |
+----------+------------+---------------+
|        1 | 2026-02-20 | Rahul Sharma  |
|        2 | 2026-02-21 | Amit Verma    |
|        3 | 2026-02-22 | Priya Singh   |
|        4 | 2026-02-23 | Neha Gupta    |
|        5 | 2026-02-24 | Rohit Kumar   |
+----------+------------+---------------+
5 rows in set (0.001 sec)
```

Q27. Display the total number of purchases made.

```
MariaDB [inventory_db]> SELECT COUNT(*) AS Total_Purchases
    -> FROM Purchase;
+-----------------+
| Total_Purchases |
+-----------------+
|               5 |
+-----------------+
1 row in set (0.001 sec)
```

Q28. Display the total quantity purchased for each product.

```
MariaDB [inventory_db]> SELECT
    ->     product_id,
    ->     SUM(quantity) AS Total_Quantity_Purchased
    -> FROM Purchase_Detail
    -> GROUP BY product_id;
+------------+-------------------------+
| product_id | Total_Quantity_Purchased |
+------------+-------------------------+
|          1 |                       5 |
|          2 |                      20 |
|          3 |                      10 |
|          4 |                       5 |
|          5 |                       5 |
+------------+-------------------------+
5 rows in set (0.001 sec)
```

Q29. Display the names of products that have never been sold.

```
MariaDB [inventory_db]> SELECT name
    -> FROM Product
    -> WHERE product_id NOT IN (
    ->     SELECT product_id
    ->     FROM Sales_Details
    -> );
Empty set (0.005 sec)
```

Q30. Display the total sales amount for each sales transaction.

```
MariaDB [inventory_db]> SELECT
    ->     sales_id,
    ->     SUM(quantity * selling_price) AS Total_Sales_Amount
    -> FROM Sales_Details
    -> GROUP BY sales_id;
+----------+--------------------+
| sales_id | Total_Sales_Amount |
+----------+--------------------+
|        1 |           52000.00 |
|        2 |            1200.00 |
|        3 |            1000.00 |
|        4 |           13000.00 |
|        5 |           17000.00 |
+----------+--------------------+
5 rows in set (0.001 sec)
```

Q31. Display distinct product categories available in the inventory.

```
MariaDB [inventory_db]> SELECT DISTINCT category
    -> FROM Product;
+-------------+
| category    |
+-------------+
| Electronics |
+-------------+
1 row in set (0.001 sec)
```

Q32. Display the total number of sales made by each customer.

```
MariaDB [inventory_db]> SELECT
    ->     customer_id,
    ->     COUNT(sales_id) AS Total_Sales
    -> FROM Sales
    -> GROUP BY customer_id;
+-------------+-------------+
| customer_id | Total_Sales |
+-------------+-------------+
|           1 |           1 |
|           2 |           1 |
|           3 |           1 |
|           4 |           1 |
|           5 |           1 |
+-------------+-------------+
5 rows in set (0.001 sec)
```

Q33. Display products whose stock quantity is greater than the average stock.

```
MariaDB [inventory_db]> SELECT name, stock_quantity
    -> FROM Product
    -> WHERE stock_quantity > (
    ->     SELECT AVG(stock_quantity)
    ->     FROM Product
    -> );
+----------+----------------+
| name     | stock_quantity |
+----------+----------------+
| Mouse    |             98 |
| Keyboard |             79 |
+----------+----------------+
2 rows in set (0.001 sec)
```

Q34. Display the total sales amount greater than 10,000 for each sales transaction.

```
MariaDB [inventory_db]> SELECT
    ->     sales_id,
    ->     SUM(quantity * selling_price) AS Total_Amount
    -> FROM Sales_Details
    -> GROUP BY sales_id
    -> HAVING SUM(quantity * selling_price) > 10000;
+----------+--------------+
| sales_id | Total_Amount |
+----------+--------------+
|        1 |     52000.00 |
|        4 |     13000.00 |
|        5 |     17000.00 |
+----------+--------------+
3 rows in set (0.001 sec)
```

Q35. Display the latest sales transaction.

```
MariaDB [inventory_db]> SELECT *
    -> FROM Sales
    -> ORDER BY sales_date DESC
    -> LIMIT 1;
+----------+------------+-------------+
| sales_id | sales_date | customer_id |
+----------+------------+-------------+
|        5 | 2026-02-24 |           5 |
+----------+------------+-------------+
1 row in set (0.001 sec)
```

Q36. Display the earliest purchase transaction.

```
MariaDB [inventory_db]> SELECT *
    -> FROM Purchase
    -> ORDER BY purchase_date ASC
    -> LIMIT 1;
+-------------+---------------+-------------+
| purchase_id | purchase_date | supplier_id |
+-------------+---------------+-------------+
|           1 | 2026-02-01    |           1 |
+-------------+---------------+-------------+
1 row in set (0.001 sec)
```

Q37. Update the price of a specific product.

```
MariaDB [inventory_db]> UPDATE Product
    -> SET price = price + 500
    -> WHERE product_id = 1;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Q38. Delete a customer record with a specific ID.

```
MariaDB [inventory_db]> DELETE FROM Customer
    -> WHERE customer_id = 5;
Query OK, 1 row affected (0.003 sec)
```

Q39. Display products along with their supplier names.

```
MariaDB [inventory_db]> SELECT
    ->     p.name AS product_name,
    ->     s.name AS supplier_name
    -> FROM Product p
    -> JOIN Supplier s
    -> ON p.supplier_id = s.supplier_id;
+--------------+------------------+
| product_name | supplier_name    |
+--------------+------------------+
| Laptop       | ABC Traders      |
| Mouse        | XYZ Distributors |
| Keyboard     | Global Supplies  |
| Monitor      | Tech Wholesale   |
| Printer      | Prime Traders    |
+--------------+------------------+
5 rows in set (0.001 sec)
```

Q40. Display total number of sales transactions made in February.

```
MariaDB [inventory_db]> SELECT COUNT(*) AS February_Sales
    -> FROM Sales
    -> WHERE MONTH(sales_date) = 2;
+----------------+
| February_Sales |
+----------------+
|              4 |
+----------------+
1 row in set (0.001 sec)
```